

TRIGONOMETRIC IDENTITIES & SPECIAL TRIANGLES

Trigonometric Functions

$$\sin \theta = \frac{y}{r} \quad \csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{x}{r} \quad \sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{y}{x} \quad \cot \theta = \frac{1}{\tan \theta}$$

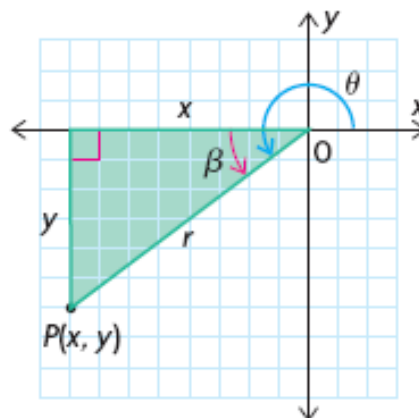
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\csc \theta = \frac{r}{y}$$

$$\sec \theta = \frac{r}{x}$$

$$\cot \theta = \frac{x}{y}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$



Equivalent Trig Expressions

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = \cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

$$\sin \theta = \sin(\pi - \theta)$$

$$\cos \theta = \cos(2\pi - \theta)$$

$$\tan \theta = \tan(\pi + \theta)$$

Cofunction Identities

$$\pm \sin \theta = \cos\left(\frac{\pi}{2} \mp \theta\right)$$

$$\cos \theta = \sin\left(\frac{\pi}{2} \mp \theta\right)$$

$$\pm \tan \theta = \cot\left(\frac{\pi}{2} \mp \theta\right)$$

Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Sum or Difference of Two Angles

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

$$\tan(a \pm b) = \frac{\tan a \pm \tan b}{1 \mp \tan a \tan b}$$

Double Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 2 \cos^2 \theta - 1$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Special Triangles

